# Anuroop Dasgupta

PhD Candidate in Astrophysics | ESO-Chile / Universidad Diego Portales

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## RESEARCH SNAPSHOT

My research asks how disks of gas and dust evolve into planetary systems, and what observable signatures trace that transformation. I am an observational astrophysicist working in high-angular-resolution astronomy, and I bring together data from millimeter to optical wavelengths:ALMA, VLT/ERIS, VLTI/GRAVITY, CHARA, and JWST:to study protoplanetary disks from their outer dust reservoirs down to sub-au inner regions. I build the reduction and modeling pipelines this work depends on, from high-contrast imaging and interferometric model fitting to time-domain survey searches for eruptive young stars. This work is grounded in hands-on observing and instrumentation: I have served as a night astronomer at La Silla and contribute to GRAVITY acquisition-camera development at Paranal, experience that shapes how I design observations and build the pipelines behind them. By combining imaging, spectroscopy, and interferometry across scales, I aim to connect the structure and evolution of disks to the companions and planets forming inside them.

## RESEARCH INTERESTS

- Protoplanetary disk structure and evolution: sizes, substructure, multiplicity, and evolutionary stage (ALMA).
- Substellar companions and early planet formation: high-contrast imaging and interferometric detection of embedded companions.
- Inner disk structure, accretion, and clearing on sub-au scales with optical/infrared interferometry (VLTI/GRAVITY, CHARA).
- Time-domain surveys and eruptive young stars: systematic searches for FU Orionis outbursts and their disks.

## EDUCATION

*March 2024 – Present*

**PhD in Astrophysics**
*European Southern Observatory & Universidad Diego Portales, Santiago, Chile*

Structure and evolution of protoplanetary systems from infrared and millimeter observations, with an emphasis on high-angular-resolution interferometric studies of young stellar objects.

Supervisors: Dr. Lucas Cieza (UDP) & Dr. Aaron Labdon (ESO).

*2021 – Nov 2023*

**M.Sc. in Physics**
*Presidency University, Kolkata, India*

Specialization in astrophysics and cosmology.

*2018 – Oct 2021*

**B.Sc. in Physics (Hons.)**
*Presidency University, Kolkata, India*

## PUBLICATIONS

### First Author 2 published · 2 to be submitted · 2 in preparation

- **Dasgupta, A.**, Labdon, A. *First K-band Interferometric Survey of Previously Unresolved Herbig Ae/Be Stars with VLTI/GRAVITY and CHARA.* To be submitted to A&A, 2026.
- **Dasgupta, A.**, Roy, N., Samui, S., Chatterjee, R., Bian, F. *Spatially Resolved Gas-Phase Metallicity Gradients in 382 MaNGA Low-Mass Galaxies: The Role of Concentration, Gas Velocity Dispersion, and Environment.* To be submitted to A&A, 2026.
- **Dasgupta, A.**, et al. *Systematic Search for FU Orionis Outbursts in ZTF: A Multi-Step Photometric Pipeline and New Candidates.* In prep.
- **Dasgupta, A.**, Cieza, L., et al. *A JWST–ALMA View of the FUor HBC 687.* In prep.
- **Dasgupta, A.**, Weber, P., Maio, F., et al. *Characterizing a New Companion Candidate Near V960 Mon.* *ApJL*, 2025. doi:10.3847/2041-8213/ade996 [ESO Press Release, July 2025]
- **Dasgupta, A.**, Cieza, L. A., Bhowmik, T., et al. *Size Distributions of Protoplanetary Disks in the Ophiuchus Molecular Cloud.* *ApJL*, 2025. doi:10.3847/2041-8213/adb03c

### Contributing Author 3 published / submitted

- Bernardi, A., Zurlo, A., Lazzoni, C., Desidera, S., Mesa, D., Pérez, S., Nogueira, P. H., Barbato, D., **Dasgupta, A.** *SaNDi-SHoP: Searching for Satellites'N'Disks with a Star-Hopping Program II. Spectrophotometric Analysis and Orbital Monitoring of Directly Imaged Companions.* Submitted to A&A.
- Lazzoni, C., Zurlo, A., Desidera, S., Bernardi, A., Pérez, S., Mesa, D., Barbato, D., Nogueira, P. H., **Dasgupta, A.**, et al. *SaNDi-SHoP: Searching for Satellites'N'Disks with a Star-Hopping Program I. arXiv*:2603.24796, 2026. doi:10.48550/arXiv.2603.24796
- Orcajo, S., Cieza, L. A., Guilera, O. M., Pérez, S., González-Ruilova, C., Batalla-Falcon, G., Bhowmik, T., Chavan, P.,

Casassus, S., **Dasgupta, A.**, et al. *ODISEA: A Unified Evolutionary Sequence of Planet-Driven Substructures Explaining the Diversity of Disk Morphologies. ApJL*, 2025. doi:10.3847/2041-8213/adcd58

## CODE & SOFTWARE

*2025 – Present*

**FU Orionis Discovery Pipeline — ZTF/ALeRCE**
*Python · end-to-end time-domain survey pipeline*

Mines the full ZTF archive (∼131 million sources, served via the ALeRCE broker) for FU Orionis accretion outbursts through a multi-stage selection cascade: detection and baseline cuts, WISE infrared colour selection of disk-bearing YSOs, amplitude and light-curve morphology criteria, and Lomb–Scargle vetting. Recovers all known ZTF-era FUors (Gaia18dvy, Gaia20bdk) and yields 245 vetted candidates, the largest uniformly selected FUor sample to date. Includes a real-time weekly monitor issuing automated alerts for Target-of-Opportunity follow-up, designed to be portable to the Rubin/LSST alert stream. (Repository to be released with the paper.)

*2025 – Present*

**Temperature Gradient Disk Model — VLTI/CHARA Interferometry**
*Python · interferometric + SED modeling*

Physically motivated disk model that jointly fits spectrally dispersed near-infrared squared visibilities and the broadband SED, computing disk visibilities via the Hankel transform of a radial temperature profile. Combines Nelder–Mead optimisation with `emcee` MCMC for parameter estimation, with halo and compact-component extensions. Used to constrain inner rim temperatures, radii, and temperature gradients across the Herbig Ae/Be survey.

*2024 – Present*

**ERIS/VLT $L'$-Band High-Contrast Imaging Pipeline**
*Python · ADI, PSF subtraction, sub-pixel astrometry*

Dedicated angular differential imaging and PSF-subtraction pipeline with precision astrometric extraction for VLT/ERIS. Applied to V960 Mon, where it yielded the detection of a new substellar companion candidate (ApJL 2025).

*2025 – Present*

**JWST MIRI MRS Reduction & Analysis Pipeline**
*Python · IFU spectral-cube reduction and diagnostics*

Custom reduction and analysis layer built on the MINDS / `jwst` / VIP framework for MIRI MRS (5–28 $\mu$m) data of FUor stars. Resolves IFU-cube recentering failures via peak-finding on median-collapsed cubes, with extinction correction, multi-blackbody and spline continuum fitting, and dust/ice and outflow diagnostics (10 $\mu$m silicates, $CO_2$ ice, [Ne II], $H_2$).

## PRESS COVERAGE

*July 2025*

**ESO Press Release eso2513 — "Astronomers witness newborn planet sculpting the dust around it"**
*European Southern Observatory | first-author result*

National/international release featuring my first-author study of V960 Mon (ApJL 2025); release imagery credited to ESO/A. Dasgupta.

*May 2025*

**ALMA Press Release — "ALMA inspires new models of how planet-forming disks evolve"**
*Atacama Large Millimeter/submillimeter Array | co-author result*

Release on the ODISEA unified evolutionary sequence of planet-driven disk substructures (Orcajo et al., ApJL 2025), on which I am a co-author.

## ACCEPTED TELESCOPE PROPOSALS

| **Period** | **Role** | **Title** |
|---|---|---|
| ESO P117 (2026) | CoI | ERIS/VLT: A Direct Test of Models Linking Disc Substructures to Planet Formation through Core Accretion |
| ESO P116 (2026) | CoI | GRAVITY/VLTI: Resolving the Inner Disks of a Complete Sample of Planet-Forming Disks |
| NTT/EFOSC2 (2026) | PI | Spectroscopic Classification of New Young Eruptive Star Candidates |
| ESO P115 (2024) | PI | VLT/VISIR: Formation and Distribution of Crystalline Material in Outbursting FUor Disks |
| ESO P115 (2024) | CoI | Searching for PDS 70 Analogues |
| ESO P114 (2024) | PI | Satellites & Circumplanetary Disks Around DI Companions via Star-Hopping (Cont.) |
| ESO P114 (2024) | CoI | First Survey of (Sub-)Stellar Companions Around Protostars (Part II) |

## CONFERENCES, TALKS & COLLOQUIA

Colloquia & Seminars: 4 · Talks & Posters: 9 (7 international, 2 national)

*Jul 2026*

**Discs on the Exe, University of Exeter**
Exeter, UK | *V960 Mon: Ideal Laboratory for Gravitational Instability (talk); First Volume-Complete Survey on Nearby Herbigs with GRAVITY (poster)*

*May 2026*

**Eruptive Young Stars Workshop, Konkoly Observatory**
Budapest, Hungary | *JWST–ALMA View of HBC 687 (talk)*

*Feb 2026*

**Spirit of Lyot 6, Caltech**

Pasadena, CA, USA | *V960 Mon: Ideal Laboratory for Gravitational Instability (poster)*

*Nov 2025*
**Stellar Variability: Taking the Pulse of the Universe, IUCAA**
Pune, India | *V960 Mon: Ideal Laboratory for Gravitational Instability (talk)*

*Sep 2025*
**Indian Institute of Technology Bombay**
Mumbai, India | *Planet Formation Across Different Wavelengths (invited)*

*Sep 2025*
**UKI Discs Meeting, University of Hertfordshire**
Hertfordshire, UK | *Complete Size Distribution for the 100 Brightest Protoplanetary Disks in Ophiuchus (talk)*

*Oct 2025*
**IISER Kolkata**
Kolkata, India | *Planet Formation Across Different Wavelengths (invited)*

*Feb 2025*
**Circumplanetary Disks and Satellite Formation III**
Kyoto, Japan | *Complete Size Distribution for the 100 Brightest Protoplanetary Disks in Ophiuchus (talk)*

*Dec 2024*
**Tata Institute of Fundamental Research (TIFR)**
Mumbai, India | *Size Distributions of Protoplanetary Disks in Nearby Star-Forming Regions (invited)*

*Dec 2024*
**National Centre for Radio Astrophysics (NCRA)**
Pune, India | *Size Distributions of Protoplanetary Disks in Nearby Star-Forming Regions (invited)*

*Nov 2024*
**SOCHIAS XIX Meeting, Universidad de Tarapacá**
Arica, Chile | *What is the Size Distribution of Disks in Nearby Star-Forming Regions? (contributed)*

*Nov 2024*
**Unveiling the Origin of Brown Dwarfs, ESO**
Santiago, Chile | *First Direct Observational Evidence of Gravitational Instability? (contributed)*

## OBSERVING & INSTRUMENTATION

*March 2026*
**Night Astronomer — NTT/EFOSC2, La Silla Observatory**
*ESO-Chile, La Silla, Chile*

Conducted service observations over 10 nights, executing 13 approved ESO-Chile programs spanning optical imaging and low/medium-resolution spectroscopy, including target acquisition, real-time instrument operation, and quality control.

*2025 – Present*
**GRAVITY Acquisition-Camera Photometry — VLTI/GRAVITY**
*ESO & UDP | instrument development*

Developing reliable photometric calibration from the GRAVITY acquisition camera in dual-field mode, identifying NIR standards suitable for dual-field observations, preparing and executing test OBs, and assessing flux-calibration stability and accuracy, toward formalising acquisition-camera photometry as a standard science capability.

*July 2025*
**Paranal Observatory — VLTI/GRAVITY Calibration**
*ESO, Atacama Desert, Chile*

Participation in VLTI operations and GRAVITY instrument calibration.

*Oct 2024*
**Swope Telescope Observing Run**
*Las Campanas Observatory, Chile*

Multi-filter imaging; project on the metallicity–dynamics connection in galaxies.

## SUPERVISION

*2025 – Present*
**MSc Thesis Co-Supervisor — Srimoyree Mitra (St. Xavier's College, Kolkata, India)**
*Universidad Diego Portales*

"Evolution of Light Curves Across SED Classes in Protoplanetary Disks in the Taurus Molecular Cloud and the Effect of Multiplicity." Multi-epoch photometric variability analysis across disk evolutionary stages.

## TECHNICAL SKILLS

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|---|---|
| **Facilities** | ALMA (Bands 6–8), VLT/ERIS, VLTI/GRAVITY, CHARA/MIRCX-MYSTIC, VLT/VISIR, JWST/MIRI, NTT/EFOSC2, Swope |
| **Programming** | Python (primary), Fortran, Mathematica, LaTeX, Bash; Git, test-driven development |
| **Astro tools** | CASA, PMOIRED, `jwst`/MINDS/VIP, Astropy, Specutils, `emcee`, IRAF, DS9, TOPCAT, Streamlit |
| **Analysis** | Interferometric visibility modeling · Bayesian inference / MCMC · Lomb–Scargle · multi-wavelength photometry (ZTF, WISE, 2MASS, Gaia, PanSTARRS) · spectral line fitting · bootstrap / Monte Carlo |
| **Pipelines** | End-to-end instrument-adaptable workflows; large-scale DB queries (SQL, ALeRCE API); automated real-time alert systems |

## AWARDS, FELLOWSHIPS & HONOURS

- **ANID Beca de Doctorado Nacional** (National Doctoral Fellowship) — Agencia Nacional de Investigación y Desarrollo (ANID), Chile, 2025.
- **ESO Studentship Programme** — European Southern Observatory, 2025–2027.
- **Outstanding Paper Award** — awarded for the best paper/presentation at the state level (West Bengal, India); 31st West Bengal State Science & Technology Congress, 2024.
- **Paramesh Chandra Bhattacharya Memorial Award** — Best Master's thesis, Presidency University, 2023.
- **Academic Ranking** — ranked 3rd in M.Sc. semesters 3 & 4, Presidency University; ranked 14th/~1,000 in Pune University MSc entrance (declined).
- **Top 1% West Bengal State Board**, Class 10 & 12 (2018); selected for INSPIRE *Young Scientists Meet*, DST India, 2016.

## OUTREACH

- Great Asteroid Festival (5th ed.) — AUI/NRAO/YEMS outreach stand, Las Vizcachas, Puente Alto, Chile (2026).
- Colloquium Host, ESO-Chile weekly colloquium programme (ongoing).
- "Exploring Exoplanets" (PME-funded, YEMS Millennium Nucleus "Astrophysics at School") — developing an educational card game and activity book on exoplanets, disks, and Solar System moons for underserved communities (ongoing).
- Public telescope observations and guided eclipse viewing (Universidad Diego Portales); science communication and online seminars introducing astrophysics to students in India and Chile.

## SCHOOLS & WORKSHOPS

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|---|---|
| Summer School in Astrophysics — Indian Institute of Astrophysics, Bangalore | *2022* |
| Summer Research Fellowship (High Energy Astrophysics) — IISER Bhopal | *2022* |
| International Capsule Workshop on Astrobiology — Indian Astrobiology Research Centre | *2019* |

## REFERENCES

**Dr. Lucas Cieza**
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**Dr. Aaron Labdon**
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European Southern Observatory
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